

PROBABILITY

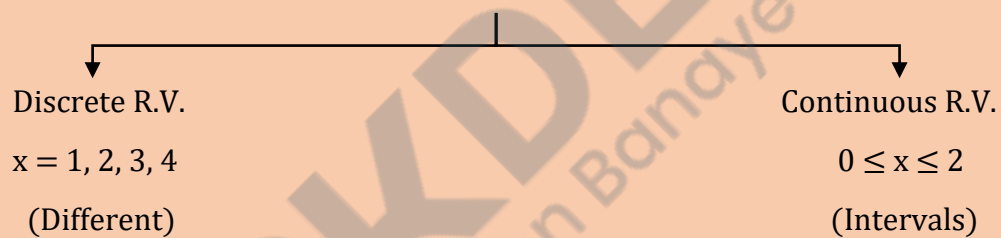
- 1) Probability Distribution
- 2) Mathematical Expectation

1) Probability Distribution

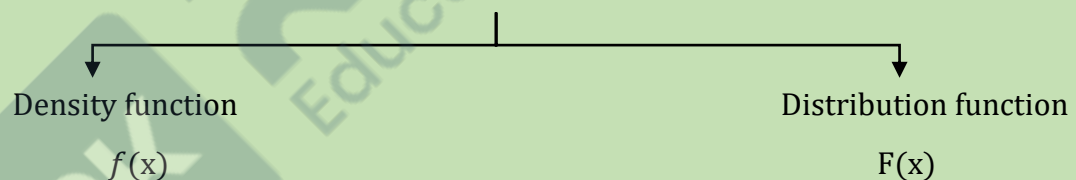
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Chances Divide

- i) Prob. is always positive
- ii) Prob. lies between 0 and 1
- iii) Addition of all prob is 1

RANDOM VARIABLE



RANDOM VARIABLE



$$f(x) = \frac{d}{dx} [F(x)]$$

PROBABILITY DENSITY/MASS FUNCTION (PDE)

Discrete R.V.

x	0	1	2	3
$p(X = x)$	0.1	0.2	0.3	0.4

$$\text{i) } \sum p(x) = 1$$

$$\text{ii) } 0 \leq p(x) \leq 1$$

Continuous R.V.

$$f(x) = 3x^2 \quad 0 \leq x \leq 1$$

$$= \int_0^1 (3x^2) dx$$

$$= 3 \left(\frac{x^3}{3} \right)_0^1$$

$$= 1 - 0$$

$$= 1$$

$$\int f(x) dx = 1$$

Distribution Function

Discrete R.V.

x	0	1	3	4
p(x)	0.1	0.2	0.3	0.4
F(x)	0.1	0.3	0.6	1

Continuous R.V.

$$F(x) = \int_{-\infty}^x f(x) dx$$

2) Mathematical Expectations

$$\text{Moment : } \mu'_r = \sum_{\text{CRV}} x^r p(x) = \int_{\text{DRV}} x^r f(x) dx$$

i) Mean: $r = 1$

$$\mu'_1 = \sum x' p(x) = \int x^1 f(x) dx$$

Mean $\rightarrow \mu'$ or $E(x)$

ii) Variance: (changes)

$$\delta^2 = v(x) = \mu'_2 - (\mu'_1)^2$$

$$= 2^{\text{nd}} \text{ moment} - (1^{\text{st}} \text{ moment})^2$$

$$[V(x) = E(x^2) - [E(x)]^2]$$

iii) Standard Deviation: (SD)

$$\delta = +\sqrt{v(x)}$$

iv) Raw Moment:

$$r = 1 : \mu'_1 = \sum x^1 P(x) = \int x^1 f(x) dx$$

$$r = 2 : \mu'_2 = \sum x^2 P(x) = \int x^2 f(x) dx$$

$$r = 3 : \mu'_3 = \sum x^3 P(x) = \int x^3 f(x) dx$$

$$r = 4 : \mu'_4 = \sum x^4 P(x) = \int x^4 f(x) dx$$

v) Central Moment:

$$\mu_1 = 0$$

$$\mu_2 = 1\mu'_2 - 1(\mu'_1)^2$$

$$\mu_3 = 1\mu'_3 - 3\mu'_2 \cdot \mu'_1 + 2(\mu'_1)^3$$

$$\mu_4 = 1\mu'_4 - 4\mu'_3 \cdot \mu'_1 + 6\mu'_2 \cdot \mu_1'^2 - 3(\mu_1')^4$$

vi) MGF: Moment Generating Function $[M_0(t)]$

$$M_0(t) = \sum e^{tx} p(x) = \int e^{tx} f(x) dx$$

$$\left[\frac{d^r}{dt^r} M_0(t) \right]_{t=0} = \mu'_R$$

BAYE'S THEOREM

$$P = \frac{p_i p'_i}{p_1 p'_1 + p_2 p'_2 + \dots \dots p_i p'_i}$$

